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PERIODIC NATURE OF ANISOTROPY IN TRANSFORMERS: HOW COMPONENTS OF TRANSFORMER LAYERS AFFECT THE GEOMETRY OF VECTORS OF TOKEN REPRESENTATIONS

Abstract. Anisotropy is the tendency of token representations to cluster along a narrow set of directions in latent space, and it is known to affect the expressive power of Transformer-based language models. While the phenomenon is well-documented at the layer level, the contribution of individual architectural components of the models remains unclear. We conduct a fine-grained component-wise analysis of seven autoregressive Transformers from the Pythia, Qwen2.5, and GPT-2 families, using cosine and singular anisotropy measures after every internal transformation. Our main finding is that anisotropy follows a periodic, component-driven pattern within each layer: anisotropy values spike after residual connections and drop after MLP and attention blocks. We further show that singular anisotropy is strongly correlated (Pearson $r = 0.91$ - 0.96) with the growing norm of the first token. Comparison with randomly initialized models confirms that the residual-driven periodicity is a learned pattern, absent before training.

§1. Introduction

Modern large language models (LLMs) based on the Transformer architecture have achieved significant success across a wide range of natural language processing tasks, such as topical summarization [28], machine translation [13], machine generated text detection [9], and coreference resolution [11]. Their internal mechanisms, particularly the patterns and dynamics of information transformation as data pass through the deep layers of the network, remain a topic of active research. One of the key phenomena characterizing the internal dynamics of Transformers is anisotropy, which quantifies the tendency of token representations to concentrate along a small number of directions in the latent space. When anisotropy is high,

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embeddings become nearly collinear, collapsing the model’s ability to distinguish contextual nuances. Empirical studies show that this collapse is not purely geometrical, but also affects the model’s perplexity. It increases the chance of repetitions in generation and reduces downstream performance [6, 24].

Understanding where exactly anisotropy arises within a model is a necessary step for addressing it. Existing research has mostly measured anisotropy at layer outputs, treating a Transformer block as a monolithic unit [4, 21], or attributed the phenomenon to isolated factors, such as weight initialization [7] or the use of tied embeddings [5, 14, 27]. However, each layer consists of several components: attention, residual addition, normalization, and MLP; all of which might affect the anisotropy in different ways. The question of the individual contribution of each of these components to the change of anisotropy dynamics throughout the model has not been systematically addressed. Answering this question could help design strategies for the mitigation of anisotropy: if a specific component is the primary source, architectural modifications or regularization can be directed precisely rather than applied broadly.

In this work, we perform a fine-grained analysis of anisotropy in seven Transformer models, including Pythia, Qwen2.5, and GPT-2 families. We extract the hidden states after every internal component within each layer and measure both cosine and singular anisotropy at each point. Our analysis provides three main findings: 1) anisotropy shows a periodic pattern within one layer: residual connections consistently increase both anisotropy measures, while attention and MLP blocks reduce them. 2) Singular anisotropy strongly correlates (Pearson $r = 0.91$ - 0.96) with relative norm of the first token in the sequence, confirming and extending the connection between massive activations and geometric collapse identified by Queipo-de-Llano et al. [18]. 3) Comparison with randomly initialized models shows that the periodic, residual-driven pattern is absent before training, indicating that it is a learned property of the optimization process rather than an architectural inevitability.

§2. Related Work

The systematic study of anisotropy in language models began with Ethayarajh et al. [4], who showed that the average pairwise cosine similarity of token vectors increases monotonically towards the output layers

of BERT, ELMo, and GPT-2. Subsequent works refined this for decoder-only models: Razzhigaev et al. [21] showed that the anisotropy profile is rather a bell-shaped curve, with a peak in the middle layers, accompanied by a two-phase change in the intrinsic dimensionality of embeddings: expansion followed by compression. Timkey et al. [24] identified that individual coordinates with disproportionately large values can inflate cosine measurements, but simple postprocessing techniques, such as standardization, partially restore representational quality.

2.1. Cause of anisotropy. The discussion of the nature of anisotropy includes several hypotheses. Noci et al. [17] linked it to weight initialization and the attention mechanism, showing theoretically that the representation matrix risks collapsing to rank 1, which leads to vanishing gradients. Then, Godey et al. [7] refined this argument, showing on MultiBERT checkpoints that the optimizer tends to shift attention distribution toward the point of maximum contrast ratio, which is geometrically achieved by aligning queries and keys along a single axis.

The use of tied embeddings can also be a causal factor: Gao et al. [5] showed that the power-law distribution of token frequencies causes gradients to “push” rare tokens into a single averaged direction. Another study [8] examines the constraint known as the “softmax bottleneck”: the mismatch between the dimension of the hidden layer and the complexity of the target distribution forces the optimizer to sacrifice representation. Diehl Martinez et al. [16] generalized all these observations by demonstrating that maximum likelihood training itself gives rise to anisotropy by causing token representations to cluster tightly in a high-dimensional cone rather than utilizing the full representational capacity.

A complementary line of research analyzes Transformers as interacting particle systems, where token representations evolve across layers according to a partial differential equation describing their collective dynamics. Under simplifying assumptions, such as shared parameters, no feedforward networks, and identity value matrices, Geshkovski et al. [6] prove that self-attention naturally drives tokens to cluster, converging either to the vertices of a convex polytope or collapsing into a low-dimensional subspace. For autoregressive models, causal masking alters the dynamics, causing tokens to converge toward the first token when the value matrix is the identity [10].

2.2. Localizing anisotropy within layers. The work closest to ours is by Skean et al. [22], who showed that the critical drop in entropy occurs specifically during the addition of the residual connection, which, according to neural ODE theory [15], suppresses high-frequency components of the update. Queipo-de-Llano et al. [18] offered a broader framework by linking the phenomena of attention sinks and compression valleys to the emergence of massive activations at the beginning-of-sequence token. The authors prove that the dominance of the first token’s norm inevitably causes the dominance of the first singular value.

2.3. Consequences and mitigation. High anisotropy negatively impacts the performance of LLMs’ and generation quality: Gao et al. [5] showed that regularization aimed at reducing anisotropy improves generation metrics. Li et al. [12] established a negative correlation between cosine anisotropy in the last layer and perplexity: stronger models distribute contextual representations more widely. Viswanathan et al. [25] connected representation geometry and model uncertainty. However, the relationship is not always straightforward: Ait-Saada and Nadif [1] found limited impact of anisotropy on text clustering tasks, and Rajaei and Pilehvar [20] showed that fine-tuning does not reliably improve isotropy.

Mitigation approaches fall into two categories. Training-time methods include regularization of pairwise cosine similarity [5], modifications to residual connections to prevent rank collapse [17]; and contrastive learning objectives, which penalizes the proximity of different token representations [23]. Post-processing methods include the technique developed by Bihani and Rayz [3], which combines the removal of principal components with subsequent semantic contraction. Our component-wise analysis suggests that mitigation efforts should be directed at residual connections, since they are the consistent source of anisotropy spikes across all models we examined.

§3. Anisotropy Measures

Each Transformer layer applies a sequence of transformations: normalization, attention, residual addition, normalization, MLP, and residual addition. We extract the matrix of token representations $\mathbf{X} \in \mathbb{R}^{n \times d}$, at each of these intermediate points, where n is the sequence length and d is the

hidden dimension. The rows $x_1, x_2, \dots, x_n \in \mathbb{R}^d$ are the individual token vectors at a given extraction point. We then evaluate two anisotropy measures on each such matrix.

The cosine anisotropy A_c is defined as the expected value of the absolute cosine similarity between two distinct rows of $\mathbf{X}$ chosen uniformly at random. Formally,

$$A_c = \mathbb{E}_{i \neq j} \left[\frac{|\mathbf{x}_i \cdot \mathbf{x}_j|}{\|\mathbf{x}_i\| \|\mathbf{x}_j\|} \right] = \frac{1}{n(n-1)} \sum_{i \neq j} \frac{|\mathbf{x}_i \cdot \mathbf{x}_j|}{\|\mathbf{x}_i\| \|\mathbf{x}_j\|}.$$

We further refer to this metric as cosine similarity. It does not take into account differences in norms between vectors and considers only the angle between them. A value A_c close to 1 indicates strong anisotropy when the vectors are nearly collinear. A value near 0 suggests isotropy when the directions are uniformly distributed.

Crucially, cosine anisotropy is not invariant to shifts: if the representation cloud drifts away from the origin, the angles between vectors shrink even if their relative arrangement is unchanged. This means that an increase in A_c does not necessarily mean directional collapse: it may indicate the growth of the mean vector norm. We return to this distinction below when motivating the use of singular anisotropy as a complementary measure.

Another variation of anisotropy. Let $\tilde{\mathbf{X}} \in \mathbb{R}^{n \times d}$ be the centered version of matrix $\mathbf{X}$, on which we compute the SVD decomposition. The singular anisotropy A_s is then defined as the ratio of the square of the largest singular value to the sum of the squares of all singular values:

$$A_s = \frac{\sigma_1^2}{\sum_{k=1}^r \sigma_k^2},$$

where r is the number of non-zero singular values. This ratio lies in $(0, 1]$ and quantifies the concentration of the vectors along the principal direction. A value close to 1 implies that most of the variance is captured by a single direction, whereas a value near $1/r$ indicates isotropy.

Because singular anisotropy is computed on centered vectors, it is invariant to shifts of the representation cloud. A large value of A_s therefore unambiguously indicates that token representations are concentrated along a single direction of variance, regardless of how far the mean vector has drifted from the origin. This makes singular anisotropy the more reliable indicator of genuine geometric collapse, while cosine anisotropy remains

useful as a signal that additionally reflects the growing dominance of the mean vector – a phenomenon that, as we show in Section 5, follows its own distinct periodic pattern.

§4. Experiments

4.1. Models. We select decoder-only models from three families that allow us to vary scale and architectural details systematically. From the Pythia family [2] we use Pythia-410m, Pythia-1.4b, and Pythia-2.8b, as these models were trained independently on the same data in the same order with identical hyperparameters, isolating model scale as the variable. From the Qwen2.5 family [26] we use Qwen2.5-0.5B, Qwen2.5-1.5B, and Qwen2.5-3B, which use RMSNorm instead of LayerNorm, allowing us to test whether normalization type affects anisotropy dynamics. We also include GPT2-XL [19] as a model trained under a different regime entirely, and whose small size makes the interpretation relatively easy.

4.2. Extraction points. To localize where anisotropy changes within each Transformer block, we extract hidden states at six fixed points of the computation graph: (i) input to the block; (ii) post-normalization state; (iii) attention output before residual addition; (iv) after the first residual addition; (v) MLP output before residual addition, and (vi) after the second residual addition (block output).

This gives paired pre/post measurements for each component: normalization, attention, residual connection, and MLP, enabling us to attribute changes in anisotropy to specific transformations. We compute both A_c and A_s at each extraction point. We first aggregate metrics at the per-sequence level and then report corpus-level means across all sampled sequences. Dispersion is reported as the standard deviation across sequences.

4.3. Dataset. The dataset we utilize for analysis is the English-language Wikipedia snapshot, accessible via the Hugging Face Datasets library as “wikimedia/wikipedia”¹ with the “20231101.en” configuration. We sampled 500 articles and truncated each to 512 tokens. Metrics are averaged over groups of 20 texts; the variance of both cosine and singular anisotropy across 25 such groups is on the order of 10^{-5} , confirming that anisotropy estimates are stable across text samples. To verify that results are not

¹<https://huggingface.co/datasets/wikimedia/wikipedia>

dataset-specific, we replicate our analysis on the RealNews subset of C4 (“allenai/c4/realnewslike”) and report those results in Section 5.

Note that all tokens in each sequence are included in the anisotropy computation, including the beginning-of-sequence token. Since the first token’s norm turns out to be strongly correlated with singular anisotropy (see Section 5), its inclusion is a measurement choice and also a feature of the analysis; therefore, excluding it would mask the very phenomenon we aim to characterize.

4.4. Baseline. As a baseline, we measured cosine and singular anisotropy in a randomly initialized model using the Pythia-1.4B architecture (i.e., without loading a pretrained checkpoint). All trainable weights were sampled from $\mathcal{N}(0, 0.02)$. We used the same tokenizer, sequence length, extraction points, and aggregation protocol as for pretrained models. This baseline isolates the contribution of learned weights: any pattern present in pretrained models but absent in the baseline can be attributed to the optimization process rather than the architecture itself.

§5. Results

5.1. Baseline. In the randomly initialized Pythia-1.4B, cosine anisotropy increases monotonically across layers, while singular anisotropy remains near zero, see Fig.1.

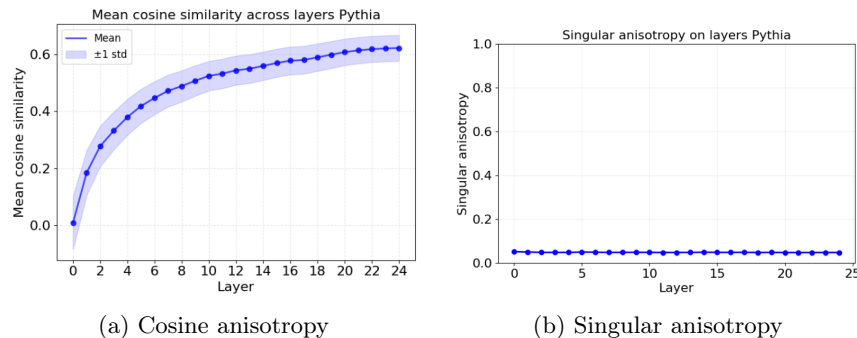


Figure 1. Layer-wise anisotropy for the randomly initialized model in which all weights were sampled from $\mathcal{N}(0, 0.02)$. Left: cosine anisotropy. Right: singular anisotropy.

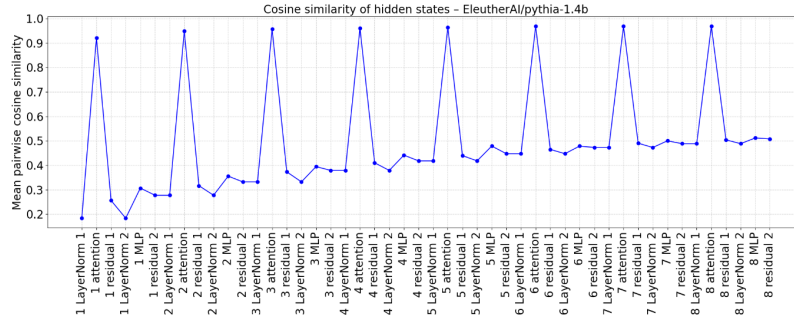
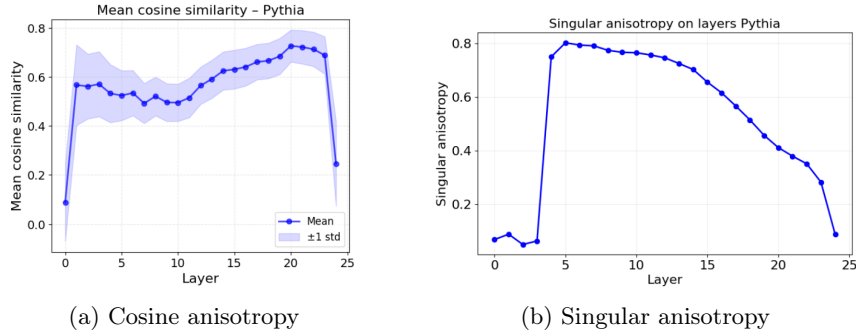


Figure 2. Component-wise cosine anisotropy on layers 1-8 for the randomly initialized model.

At the component level, the monotonic increase in cosine anisotropy is driven entirely by the attention mechanism, as each attention block raises A_c , and subsequent components do not recover it, as shown in Fig.2. The near-zero singular anisotropy indicates that the centered representations retain no dominant direction of variance. Meanwhile, the rising cosine anisotropy reflects the mean vector $\bar{\mathbf{x}}$ growing in norm, shifting the representation cloud away from the origin without compressing its internal structure. This baseline establishes two reference points: the monotonic cosine increase and the absence of periodic structure in singular anisotropy, both of which differ qualitatively from the pretrained models.

5.2. Layer-level trends in pretrained models. Across all pretrained models from Pythia and Qwen2.5 families, cosine anisotropy A_c generally increases through the layers, as shown in Fig.3a. At the same time, singular anisotropy A_s follows a different trajectory: it reaches a maximum in the early layers and then decreases to the output layer, see Fig.3b.

5.3. Component-level periodicity in pretrained models. The central finding of this work is that both anisotropy measures exhibit a periodic pattern at the component level, repeating within each Transformer block. For cosine anisotropy, values spike after residual connections and drop after attention, MLP, and generally after normalization; see Fig.4, 5. Unlike the baseline, where cosine anisotropy drifts monotonically upward with peaks only after attention, the pretrained models show a stable oscillation: residual additions raise A_c , and the subsequent transformations (normalization,



(a) Cosine anisotropy

(b) Singular anisotropy

Figure 3. Anisotropy values after each layer of Pythia-1.4b.

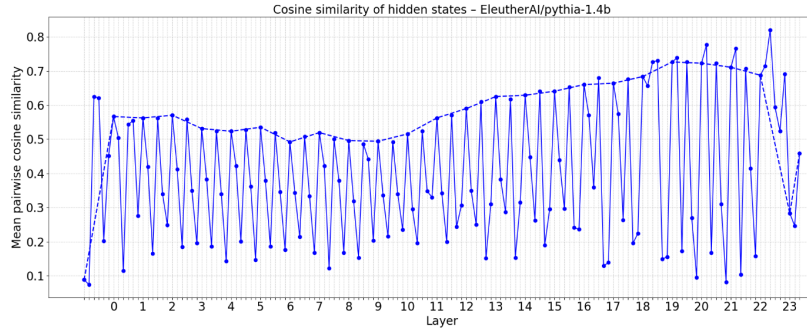


Figure 4. The periodicity of the cosine anisotropy after each layer transformation in Pythia-1.4b. The dashed line connects anisotropy values at each layer output.

attention, or MLP) bring it back down. This oscillation begins in the earliest layers and persists to the last, with values remaining below 0.8. As for singular anisotropy, the periodic structure is observed only starting from the 3rd layer; before that, it is very small, which can be attributed to vectors still not being “aligned” with each other. Once the periodicity appears, it follows the same pattern as cosine anisotropy: residual connections spike A_s , and attention and MLP reduce it (Fig.6, 7.)

The consistency of this pattern across model families is notable. In the Qwen models, which use RMSNorm instead of LayerNorm, the periodicity is preserved but with higher overall singular anisotropy values (Fig.8, 9).

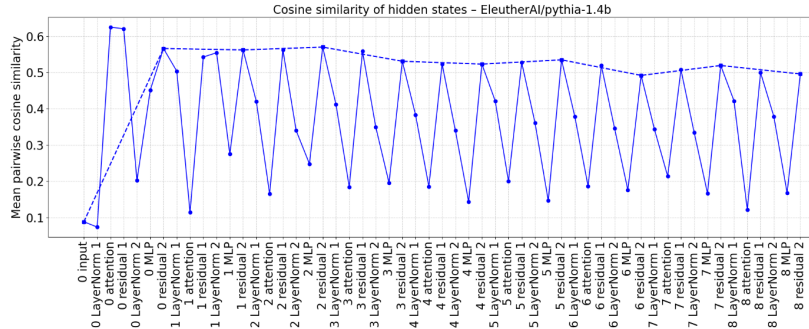


Figure 5. Cosine anisotropy after each transformation of 1-8 layers in Pythia-1.4b.

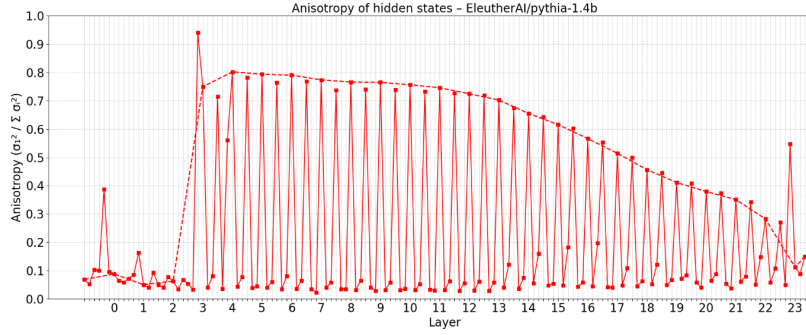


Figure 6. The periodicity of the singular anisotropy after each layer transformation in Pythia-1.4b. The dashed line connects anisotropy values at each layer output.

In GPT2-XL, cosine anisotropy exhibits a decreasing rather than increasing trend across layers, yet the within-layer periodicity with peaks after residual connections remains (Fig.10,11). The overall trends differ across families; the periodicity does not.

5.4. Component-level statistics in pretrained models. We also report the minimum, maximum, and mean values for both cosine and singular anisotropy, broken down by component type for all models on the Wikipedia dataset in Tables 1, 2 and 3. Three patterns hold across every model. First, the highest mean values of both A_c and A_s occur in residual connections. Second, the maximum A_c values per layer are reached after attention, while the

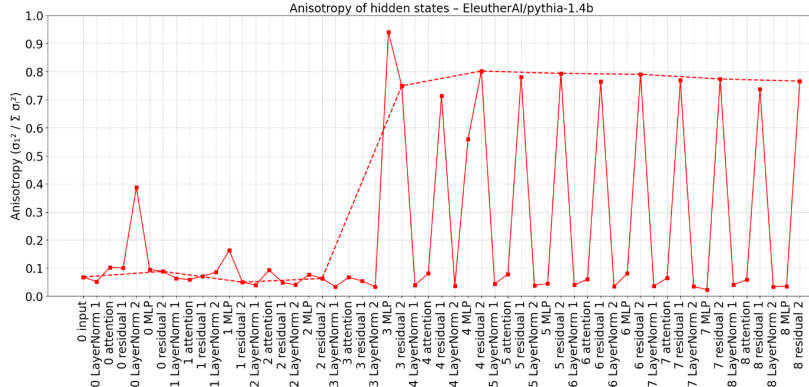


Figure 7. Singular anisotropy after each transformation on layers 1-8 in Pythia-1.4b.

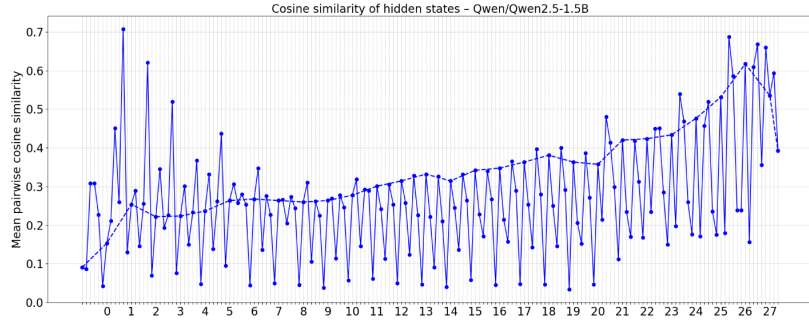


Figure 8. Cosine anisotropy after each transformation in Qwen2.5-1.5b.

maximum A_s values are reached after MLP. Third, mean cosine anisotropy increases with model scale within each family.

Both measures show a wide range of values across layers, even for the same component type. For example, in Pythia-1.4b residual connections produce values of A_s from 0.06 to 0.79 (see Fig.7), indicating that depth is affecting the component’s effect.

5.5. First-token norm. The phenomenon of singular anisotropy was linked with the emergence of massive activations on the first token in the sequence [18]. The relative norm of the first token is computed as the ratio of

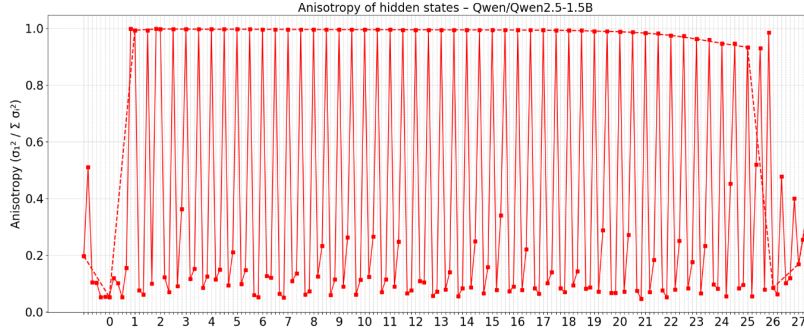


Figure 9. Singular anisotropy after each transformation in Qwen2.5-1.5b.

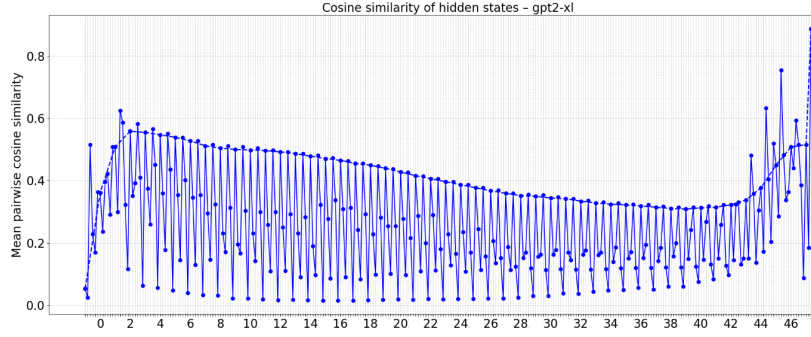


Figure 10. Cosine anisotropy after each transformation in GPT2-xl.

the first token’s norm to the sum of all token norms, and we observe that it exhibits its own periodic dynamic that closely tracks singular anisotropy (Fig.12). The relative norm of the first token spikes after MLP in the 3rd layer and has a periodic dynamic after it. Across all seven models, the Pearson correlation between the first token’s relative norm and singular anisotropy lies between 0.91 and 0.96. The correlation with cosine anisotropy is below 0.5.

This asymmetry is expected, given the definitions of metrics. When a first token dominates the norm, centering produces a covariance matrix dominated by that token’s deviation from the mean, which inflates σ_1 and therefore A_s . Cosine anisotropy, which depends on pairwise angular

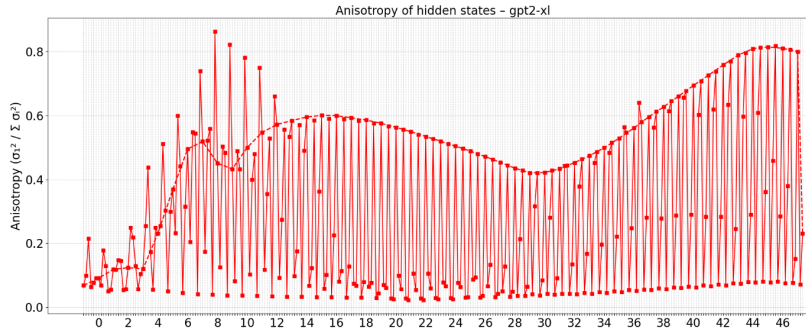


Figure 11. Singular anisotropy after each transformation in GPT2-xl.

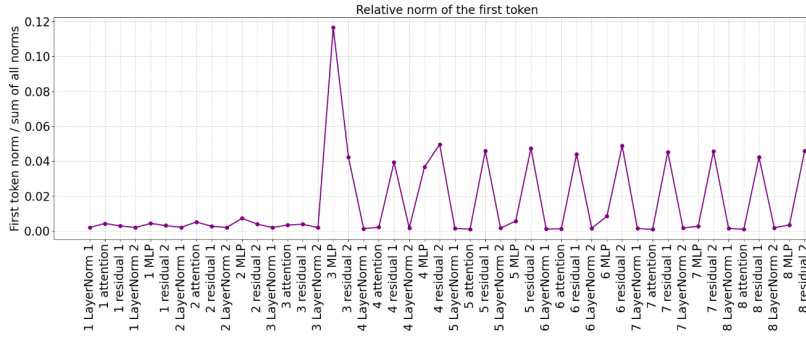


Figure 12. First token relative norm on layers 1-8 in Pythia-1.4b.

alignment across all tokens, is not mechanically tied to any single token’s norm.

5.6. Cross-dataset validation. To confirm that these patterns are not specific to Wikipedia dataset, we replicate the analysis on the RealNews² subset of C4 with English news texts. Tables 4, 5, and 6 report component-level statistics for models from Pythia and Qwen2.5 families, together with GPT-2 XL model, on this dataset. The dynamics are nearly identical: RMSE between the layer-averaged cosine anisotropy on Wikipedia and RealNews is 0.02, and for singular anisotropy it is 0.03.

²<https://huggingface.co/datasets/allenai/c4/realnewslike>

Model	Component	max A_c	min A_c	$\bar{A}_c$	max A_s	min A_s	$\bar{A}_s$
410m	Attention	0.61	0.11	0.23	0.86	0.06	0.18
	MLP	0.53	0.13	0.27	0.91	0.03	0.27
	LayerNorm	0.43	0.10	0.29	0.34	0.04	0.11
	Residuals	0.47	0.30	0.36	0.89	0.06	0.61
1.4b	Attention	0.82	0.11	0.28	0.25	0.06	0.11
	MLP	0.68	0.09	0.24	0.94	0.03	0.14
	LayerNorm	0.63	0.14	0.39	0.22	0.03	0.06
	Residuals	0.74	0.45	0.59	0.79	0.06	0.52
2.8b	Attention	0.89	0.10	0.28	0.26	0.06	0.12
	MLP	0.74	0.10	0.28	0.91	0.03	0.16
	LayerNorm	0.72	0.16	0.45	0.19	0.03	0.05
	Residuals	0.76	0.42	0.65	0.78	0.06	0.51

Table 1. Anisotropy statistics for Pythia models on Wikipedia texts

Model	Component	max A_c	min A_c	$\bar{A}_c$	max A_s	min A_s	$\bar{A}_s$
0.5B	Attention	0.70	0.09	0.28	0.58	0.06	0.17
	MLP	0.60	0.04	0.12	0.99	0.04	0.19
	LayerNorm	0.37	0.21	0.29	0.29	0.06	0.09
	Residuals	0.54	0.24	0.34	0.99	0.06	0.77
1.5B	Attention	0.68	0.09	0.24	0.46	0.06	0.20
	MLP	0.65	0.04	0.10	0.99	0.04	0.20
	LayerNorm	0.46	0.17	0.28	0.29	0.06	0.09
	Residuals	0.61	0.22	0.34	0.99	0.08	0.89
3B	Attention	0.76	0.05	0.23	0.47	0.05	0.18
	MLP	0.58	0.03	0.11	0.99	0.03	0.18
	LayerNorm	0.58	0.13	0.30	0.29	0.05	0.08
	Residuals	0.62	0.25	0.35	0.98	0.09	0.77

Table 2. Anisotropy statistics for Qwen2.5 models on Wikipedia texts

Component	max A_c	min A_c	$\bar{A}_c$	max A_s	min A_s	$\bar{A}_s$
Attention	0.75	0.08	0.21	0.64	0.05	0.31
MLP	0.52	0.02	0.08	0.86	0.02	0.24
LayerNorm	0.42	0.10	0.22	0.16	0.03	0.06
Residuals	0.58	0.30	0.42	0.80	0.08	0.51

Table 3. Anisotropy statistics for gpt2-xl on Wikipedia texts

Model	Component	max A_c	min A_c	$\bar{A}_c$	max A_s	min A_s	$\bar{A}_s$
410m	Attention	0.60	0.07	0.20	0.88	0.06	0.15
	MLP	0.58	0.12	0.27	0.91	0.02	0.27
	LayerNorm	0.46	0.11	0.30	0.30	0.04	0.09
	Residuals	0.49	0.29	0.37	0.90	0.05	0.61
1.4b	Attention	0.82	0.09	0.27	0.20	0.05	0.08
	MLP	0.75	0.10	0.25	0.94	0.02	0.14
	LayerNorm	0.65	0.15	0.40	0.22	0.02	0.05
	Residuals	0.75	0.46	0.61	0.78	0.05	0.51
2.8b	Attention	0.89	0.07	0.27	0.29	0.05	0.09
	MLP	0.81	0.10	0.30	0.90	0.02	0.15
	LayerNorm	0.72	0.16	0.46	0.19	0.02	0.03
	Residuals	0.78	0.43	0.67	0.77	0.05	0.48

Table 4. Anisotropy statistics for Pythia models on RealNews

§6. Discussion

Our central finding is that anisotropy spikes after residual connections and drops after attention, MLP, and generally after normalization. This can be explained mechanistically. Residual addition computes $x \rightarrow f(x) + x$, where f is an Attention or MLP transformation. Regardless of the directional relationship between $f(x)$ and x , the addition generally increases representation norms across layers in these models. This norm growth pushes the representation cloud further from the origin, raising cosine anisotropy through the shift mechanism described in Section 3. The mechanism behind the singular anisotropy spikes is less clear. Singular

Model	Component	max A_c	min A_c	$\bar{A}_c$	max A_s	min A_s	$\bar{A}_s$
0.5B	Attention	0.71	0.07	0.26	0.55	0.05	0.15
	MLP	0.74	0.02	0.12	0.99	0.03	0.17
	LayerNorm	0.38	0.20	0.29	0.26	0.05	0.09
	Residuals	0.54	0.22	0.33	0.99	0.05	0.77
1.5B	Attention	0.76	0.06	0.23	0.50	0.04	0.18
	MLP	0.76	0.02	0.09	1.00	0.03	0.19
	LayerNorm	0.46	0.15	0.26	0.19	0.05	0.09
	Residuals	0.63	0.22	0.34	1.00	0.06	0.89
3B	Attention	0.79	0.03	0.21	0.41	0.03	0.17
	MLP	0.70	0.02	0.11	0.99	0.02	0.16
	LayerNorm	0.57	0.10	0.30	0.29	0.04	0.07
	Residuals	0.64	0.25	0.34	0.98	0.06	0.75

Table 5. Anisotropy statistics for Qwen2.5 models on RealNews

Component	max A_c	min A_c	$\bar{A}_c$	max A_s	min A_s	$\bar{A}_s$
Attention	0.73	0.06	0.20	0.60	0.03	0.24
MLP	0.49	0.01	0.07	0.87	0.01	0.21
LayerNorm	0.44	0.10	0.22	0.15	0.02	0.05
Residuals	0.61	0.30	0.44	0.74	0.07	0.52

Table 6. Anisotropy statistics for gpt2-xl on RealNews

anisotropy increases when the variance of the centered representations becomes more concentrated along a single direction. Our data show that this consistently happens after residual addition, but we do not decompose the sub-layer outputs $f(x)$ to determine whether they reinforce the existing principal direction or introduce a new dominant one. Establishing the geometric mechanism behind the singular anisotropy spikes, for instance, by tracking the alignment between sub-layer outputs and the principal singular vector before and after addition, is a natural direction for future work.

The attention and MLP blocks act in the opposite direction, reducing both anisotropy measures. This is consistent with their role as learned

transformations that redistribute variance across dimensions, partially counteracting the compression introduced by residual addition. Normalization also generally reduces anisotropy metrics because both RMSNorm and LayerNorm affect norms and alter directions via learnable elementwise scaling; LayerNorm also shifts the mean vector.

6.1. The role of normalization. The Pythia models use LayerNorm while the Qwen models use RMSNorm, yet the periodic pattern is preserved across both families. However, Qwen models exhibit substantially higher singular anisotropy overall (Tables 1 and 2). One possible explanation is that RMSNorm, which does not subtract the mean, preserves the directional bias introduced by residual connections more than LayerNorm does. We note this as a hypothesis rather than a conclusion, and a controlled experiment varying only the normalization type while holding other factors fixed would be needed to establish causality.

6.2. First-token norm and singular anisotropy. The strong correlation (Pearson $r = 0.91$ - 0.96) between the first token’s relative norm and singular anisotropy confirms the framework of Queipo-de-Llano et al. [18]. Our contribution is showing that this relationship exhibits the same within-layer periodicity as the anisotropy measures themselves. However, the correlation is partly mechanical: when one vector dominates the norm, centering produces a covariance dominated by that vector’s deviation, inflating σ_1 by construction. The empirical finding is not the correlation itself but its periodic structure and the fact that the first token’s norm grows specifically after MLP starting in the 3rd layer, suggesting a learned computational strategy.

6.3. Limitations. Several limitations of our analysis should be noted. All models examined are decoder-only and relatively small (up to 3B parameters); whether the periodic pattern persists in larger models or in encoder-only and encoder-decoder architectures is unknown. Our analysis is observational, as we identify which components change anisotropy but do not perform ablations that would establish causal mechanisms. The first-token norm correlation, while striking, is partly a mathematical consequence of the singular anisotropy definition, as discussed above. Finally, our analysis covers English text only; whether the patterns hold for other languages or modalities is an open question.

6.4. Mitigation. Our results suggest that mitigation efforts could be targeted more precisely than current approaches allow. Since residual connections are the consistent source of anisotropy spikes, modifications to the residual pathway, such as learnable scaling of the residual branch, as explored in different contexts [17] may be more effective than regularization applied uniformly across all components. We also note that the residual-driven periodicity is learned during optimization, therefore, training-time interventions such as regularization or contrastive objectives can potentially reshape it without architectural changes.

§7. Conclusion

We conducted a component-wise analysis of anisotropy in autoregressive Transformers by extracting hidden states after every internal transformation within each layer. We found a consistent periodic pattern across three model families, two datasets, and two complementary anisotropy measures, namely, values spike after residual connections but drop after attention and MLP blocks. Both cosine and singular anisotropy are essential for analysis, as cosine anisotropy captures the drift of the representation cloud away from the origin through norm growth, and singular anisotropy captures genuine variance concentration tied to the first token’s growing relative norm. The two measures respond to the same components but reflect geometrically distinct phenomena. Comparison with a randomly initialized baseline showed that the residual-driven periodicity is learned during training, implying that training-time interventions in residual connections may be more effective than broadly applied regularization.

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